



Collections Utility Class Problem Bank (Java)

Legend

★ = Core Placement Problem (Mandatory)

Covers static utility methods provided by java.util.Collections for operating on List, Set, and Map implementations.

These problems should be completed by every student before moving to OOPs.

Section 1: Sorting and Ordering Utilities

- ★ Sort a List using `Collections.sort()`
- ★ Reverse a List using `Collections.reverse()`
- ★ Sort a List using `Collections.sort()` with a Custom Comparator
 - Shuffle a List using `Collections.shuffle()`
- ★ Rotate a List using `Collections.rotate()`
 - Swap Two Elements using `Collections.swap()`

Section 2: Searching and Frequency Utilities

- ★ Binary Search on a Sorted List using `Collections.binarySearch()`
- ★ Count Frequency of an Element using `Collections.frequency()`
 - Check if Two Collections Share No Elements using `Collections.disjoint()`

Section 3: Min/Max and Aggregation Utilities

- ★ Find Maximum Element using `Collections.max()`
- ★ Find Minimum Element using `Collections.min()`
- ★ Find Max/Min in a List of Custom Objects using a Comparator

Section 4: Immutable and Synchronized Wrappers

- ★ Create an Unmodifiable List using `Collections.unmodifiableList()`
 - Create an Unmodifiable Set or Map
- ★ Create a Thread-Safe List using `Collections.synchronizedList()`
- ★ Demonstrate `UnsupportedOperationException` on an Unmodifiable Collection

Section 5: Utility Creation Methods

- ★ Create Empty Collections using `Collections.emptyList()/emptySet()/emptyMap()`
 - Create a Singleton List using `Collections.singletonList()`
 - Create a List with N Copies of an Element using `Collections.nCopies()`



Section 6: Collection Manipulation Utilities

★ Fill a List with a Given Value using `Collections.fill()`

Copy Elements from One List to Another using `Collections.copy()`

Add Multiple Elements using `Collections.addAll()`

Replace All Occurrences of an Element using `Collections.replaceAll()`

Section 7: Real World Applications using Collections Utility Class

★ Leaderboard Sorting using `Collections.sort()` with Comparator

★ Read-Only Configuration List using Unmodifiable Collections

Randomized Quiz Question Order using `Collections.shuffle()`

★ Duplicate Vote Counter using `Collections.frequency()`